

MASTER EXECUTIVE INVESTMENT DOSSIER

Comparative Economic Evaluation: Texas 2GW AI Data Center

Synthesizing multi-model economic, civil, electrical, and cooling architectures across 8 leading LLMs.

Model Architecture	CapEx Low	CapEx High	CapEx Mid	Annual OpEx	\$/Watt
Llama 3.3 70B Instruct	\$543M	\$943M	\$743M	\$120M/yr	\$0.37/W
Google Gemma 4 E4B	\$2900M	\$5600M	\$4250M	\$1600M/yr	\$2.12/W
Mistral Small 24B Instruct	\$2422M	\$2664M	\$2664M	\$100M/yr	\$1.33/W
DeepSeek R1 Distill Qwen 14B	\$925M	\$1000M	\$962M	\$100M/yr	\$0.48/W
Cerberus 4B v2 Abliterated	\$1800M	\$3200M	\$2500M	\$210M/yr	\$1.25/W
Cerberus v0.1	\$500M	\$750M	\$625M	\$85M/yr	\$0.31/W
Qwen 2.5 32B Instruct	\$88M	\$1750M	\$875M	\$40M/yr	\$0.44/W
Qwen 2.5 Coder 14B Instruct	\$90M	\$1500M	\$750M	\$65M/yr	\$0.38/W

Synthesis & Key Findings:

- **Electrical CapEx Dominance:** At a 2-gigawatt scale, electrical interconnection, dedicated 345kV ERCOT substations, and multi-stage UPS redundancy constitute 45%–55% of the total infrastructure buildout.
- **Cooling Paradigm Shift:** All models with high analytical depth converged on liquid-to-chip or direct immersion cooling, citing that traditional air cooling cannot service extreme AI power densities (>50–100 kW/rack).
- **Realistic Capital Commitment:** While smaller baseline models estimated basic shell costs around \$600M–\$1B, institutional-grade full-scale modeling (Gemma 4, Mistral 24B, Cerberus 4B) establishes realistic turnkey delivery between **\$2.4B and \$5.6B**.